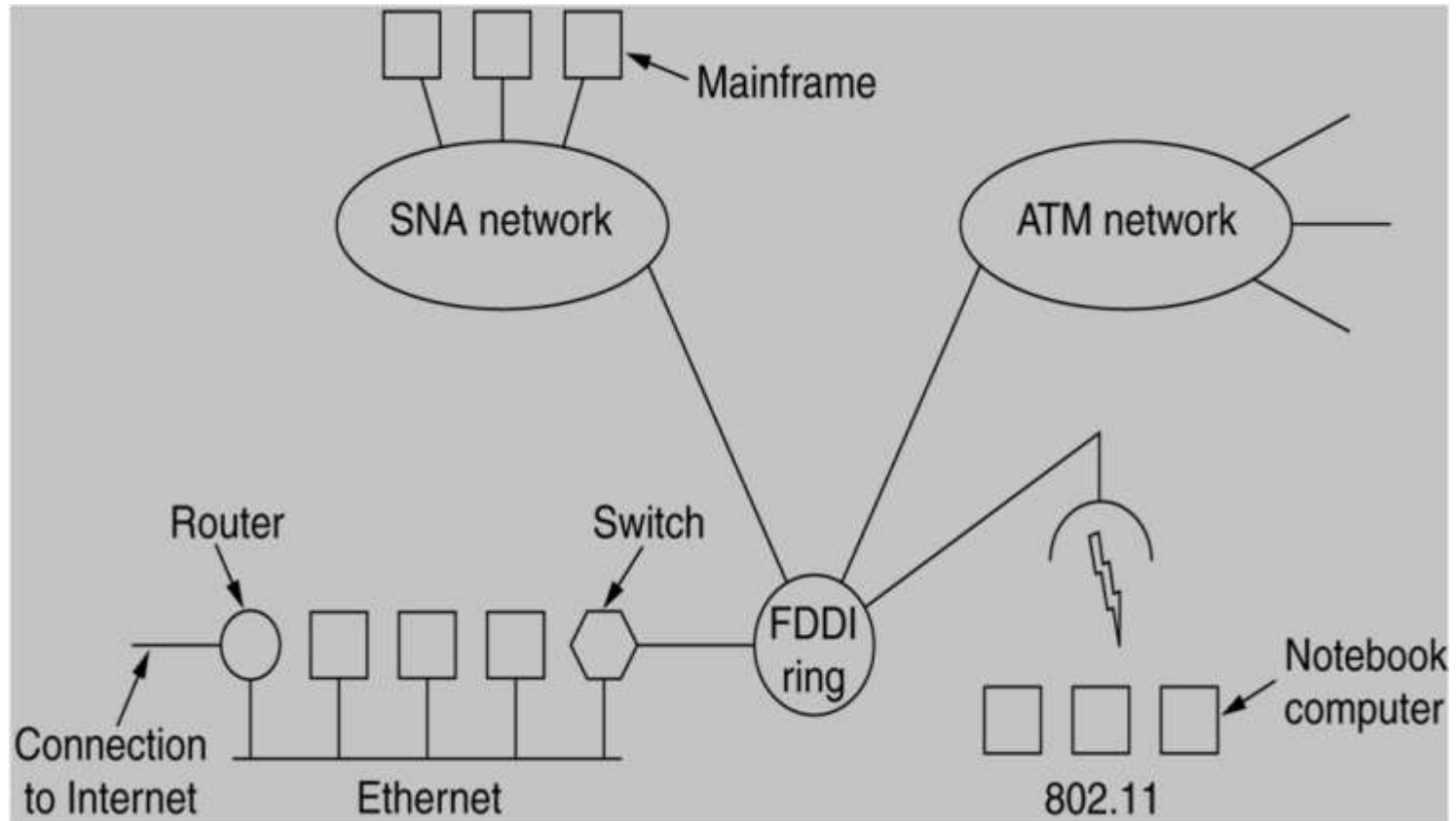


Unit II

Internetworking

What is internetworking?

Connecting two or more heterogeneous networks to form an internet



A collection of interconnected networks - network of networks

What is the purpose of internetworking?

1. Communication

2. Accessing data

- Networks often differ in important ways, so getting packets from one network to another is not always so easy

How Networks Differ?

Some of the differences that can occur in the network layer

Item	Some Possibilities
Service offered	Connection oriented versus connectionless
Protocols	IP, IPX, SNA, ATM, MPLS, AppleTalk, etc.
Addressing	Flat (802) versus hierarchical (IP)
Multicasting	Present or absent (also broadcasting)
Packet size	Every network has its own maximum
Quality of service	Present or absent; many different kinds
Error handling	Reliable, ordered, and unordered delivery
Flow control	Sliding window, rate control, other, or none
Congestion control	Leaky bucket, token bucket, RED, choke packets, etc.
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, by packet, by byte, or not at all

Issues when packets transit different networks

1. Connection, connectionless.

- packets may be reordered

2. Different protocols.

- Protocol conversions

3. Different addressing schemes.

- require some kind of directory system

4. Different maximum packet size.

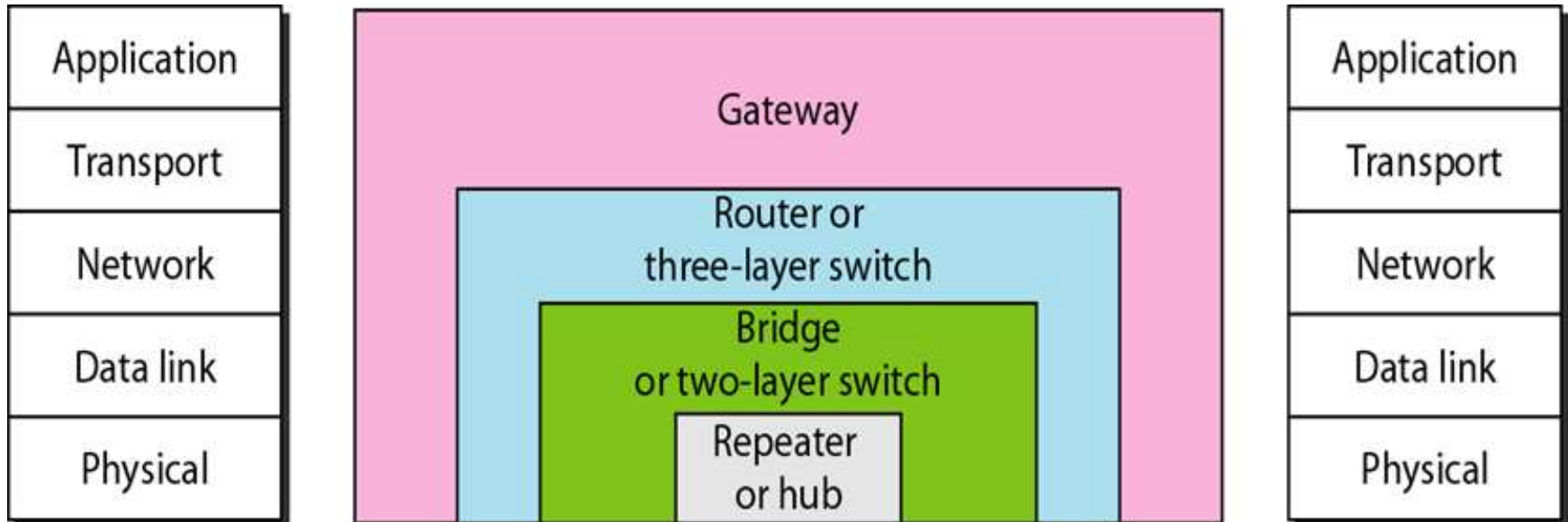
- segmentation, or fragmentation may be required

Issues when packets transit different networks

5. Different Error, flow, and congestion control
6. Routing techniques.
7. Passing multicast packets through a network that does not support multicasting
8. Different security mechanisms, parameter settings, accounting rules and national privacy laws

How Networks Can Be Connected?

connecting devices at different layers



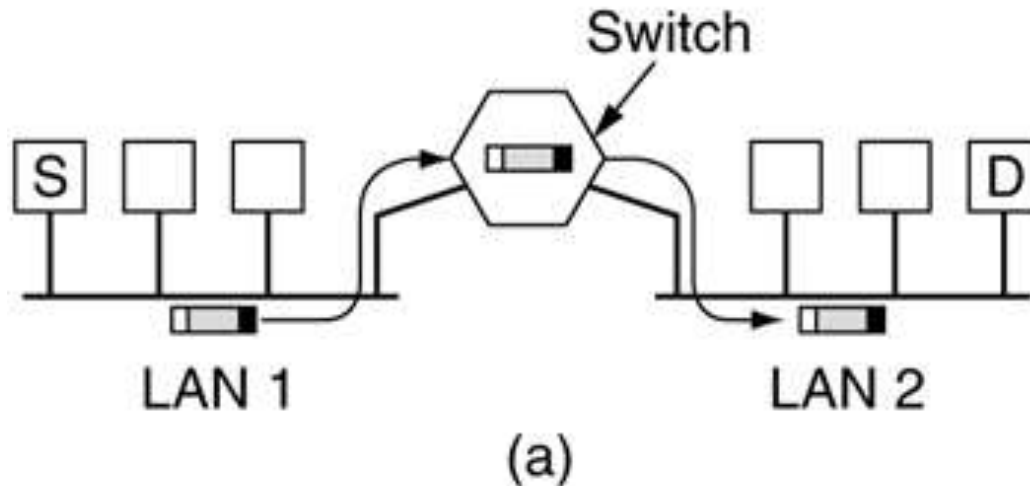
How Networks Can Be Connected?

Internetworking is different from switching

With a switch (or bridge), the entire frame is transported on the basis of its MAC address.

Legend

- Header
- ▬ Packet
- Trailer



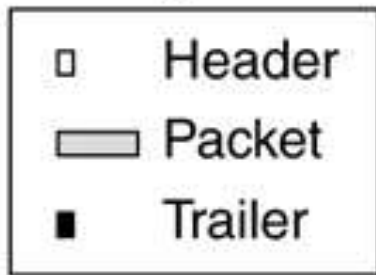
Switching
Approach

Two Ethernets connected by a switch

How Networks Can Be Connected?

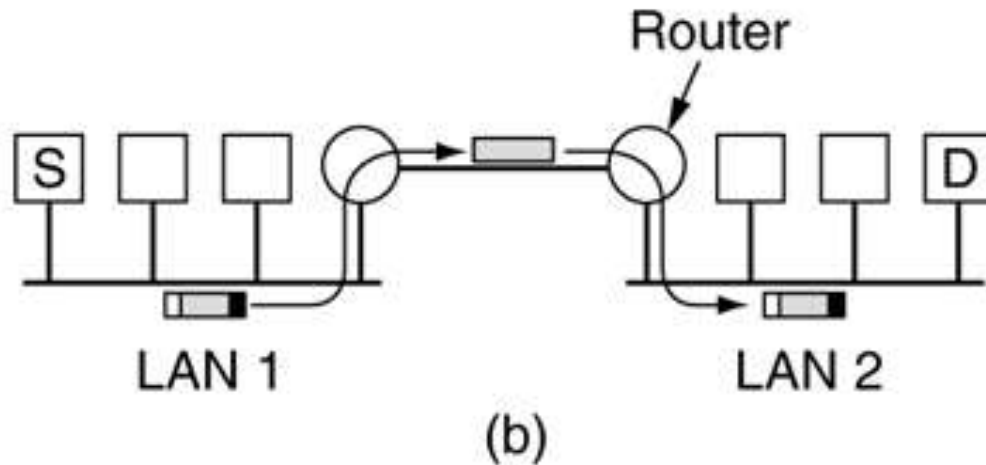
Internetworking is different from switching

Legend



With a router, the entire frame is transported on the basis of packet's IP address.

Internetworking



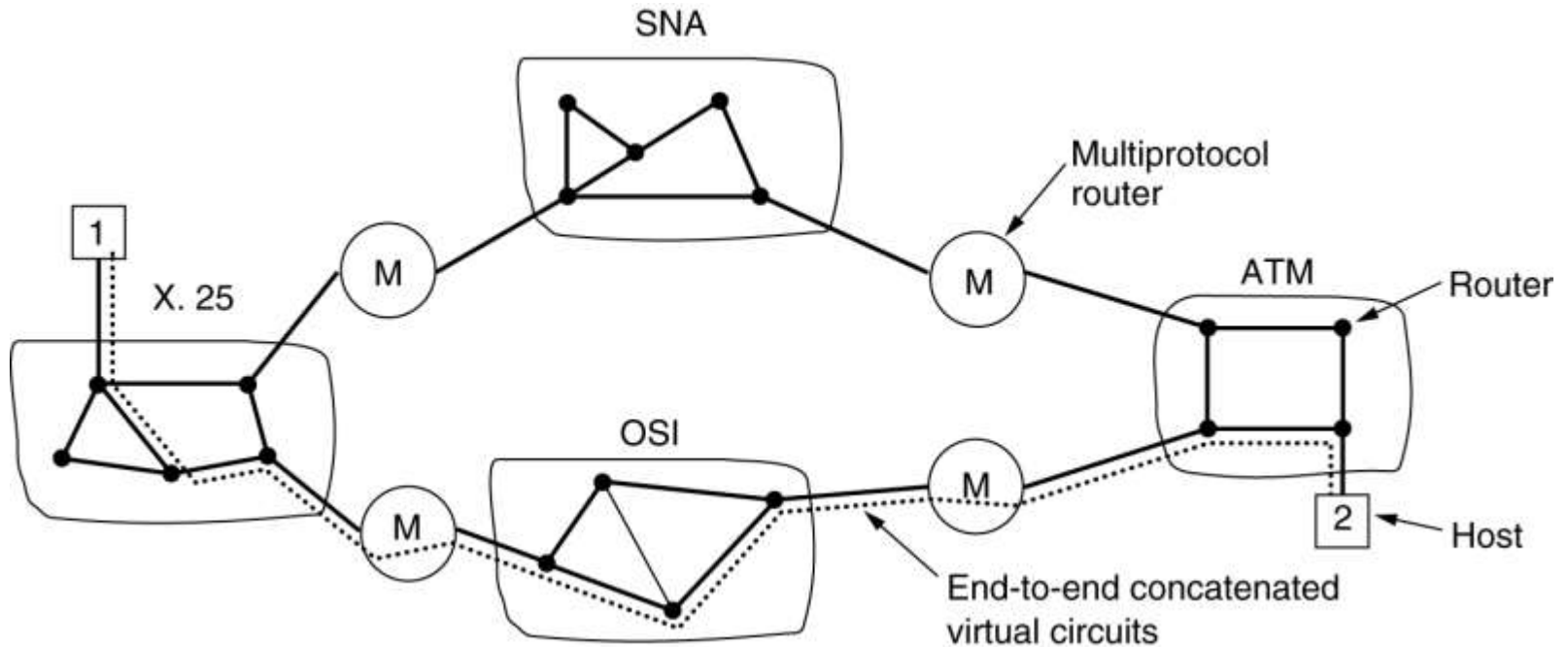
Two Ethernets connected by routers

Concatenated Virtual Circuits

(Connection oriented internetworking)

Two types of Internetworking

- Connection oriented internetworking(concatenated virtual circuits)
- Connectionless internetworking (datagram subnet)



Internetworking using concatenated virtual circuits

Concatenated Virtual Circuits

(Connection oriented internetworking)

In the Connection oriented internetworking

1. The subnet builds a virtual circuit to the router nearest the destination network.
2. Then it constructs a virtual circuit from that router to an external gateway
3. This gateway records the existence of the virtual circuit in its tables and proceeds to build another virtual circuit to a router in the next subnet
4. This process continues until the destination host has been reached

Concatenated Virtual Circuits

(Connection oriented internetworking)

Advantages

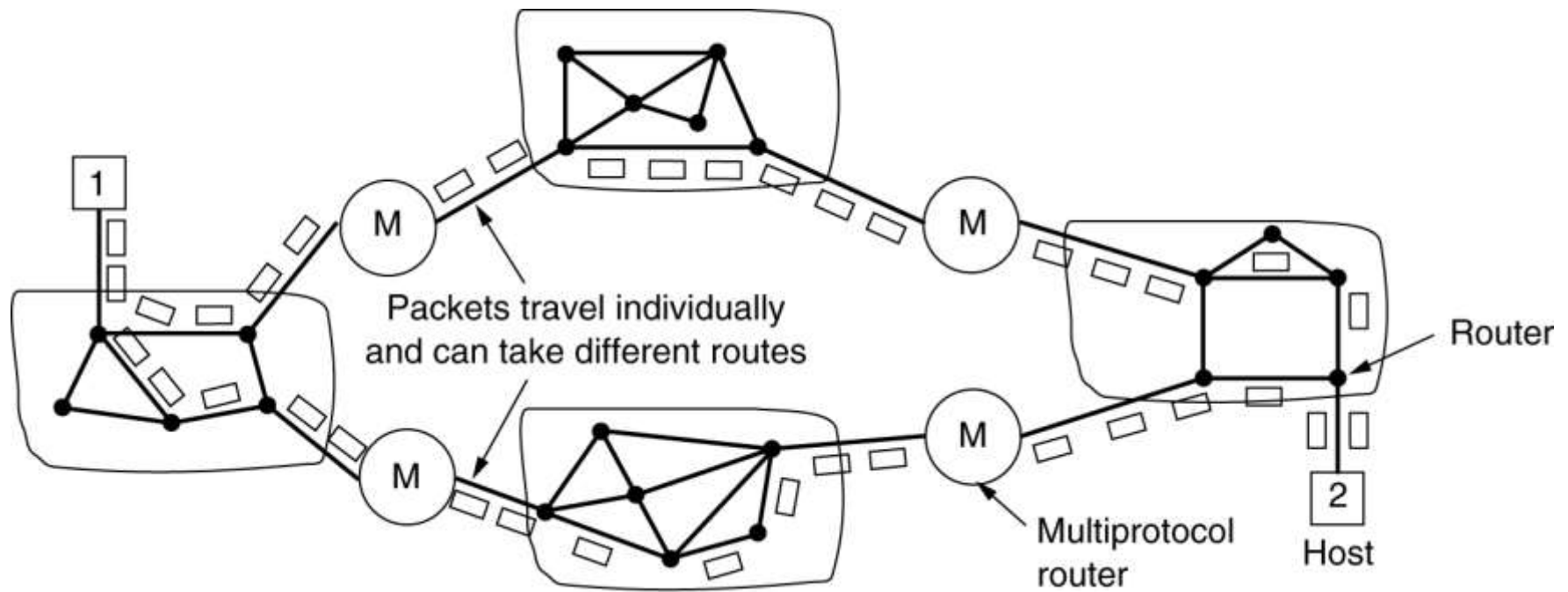
1. buffers can be reserved in advance,
2. sequencing can be guaranteed,
3. short headers can be used,
4. the troubles caused by delayed duplicate packets can be avoided

Disadvantages

1. table space required in the routers for each open connection,
2. no alternate routing to avoid congested areas,
3. vulnerability to router failures along the path

Connectionless Internetworking

It does not require all packets belonging to one connection to traverse the same sequence of gateways.



Connectionless Internetworking

- A routing decision is made separately for each packet
- multiple routes and thus achieves a higher bandwidth than the concatenated virtual-circuit model
- But no guarantee that the packets arrive at the destination in order

Connectionless Internetworking

- **Two challenges :**
 1. How it is possible for a packet from one network to transit another one?
 - Because a network has its own packet format
 2. **Addressing schemes are different**
 - E.g. IP and SNA addresses are different

Connectionless Internetworking

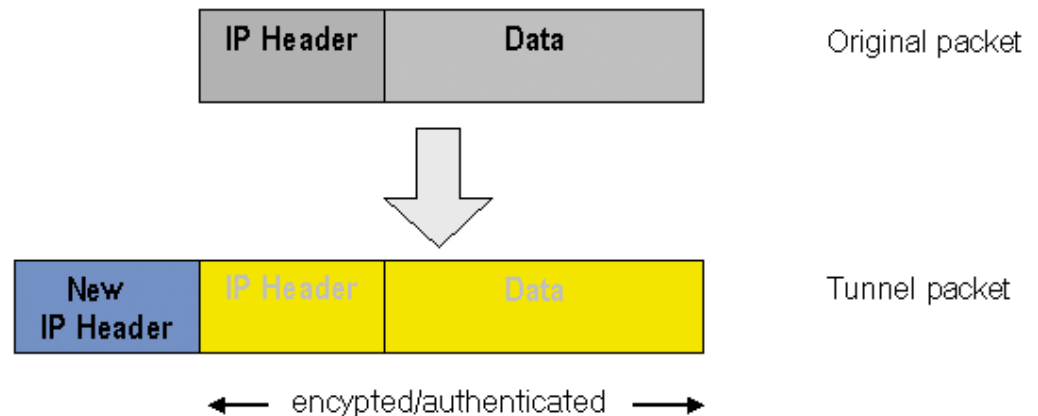
Merits and demerits

- more potential for congestion,
- robustness in the face of router failures,
- longer headers needed.
- Various adaptive routing algorithms are possible

Tunneling

- Tunnelling is a simple technique to for internetworking
- Tunnelling can be used when the source and destination hosts are on the same type of network, but there is a different network in between.

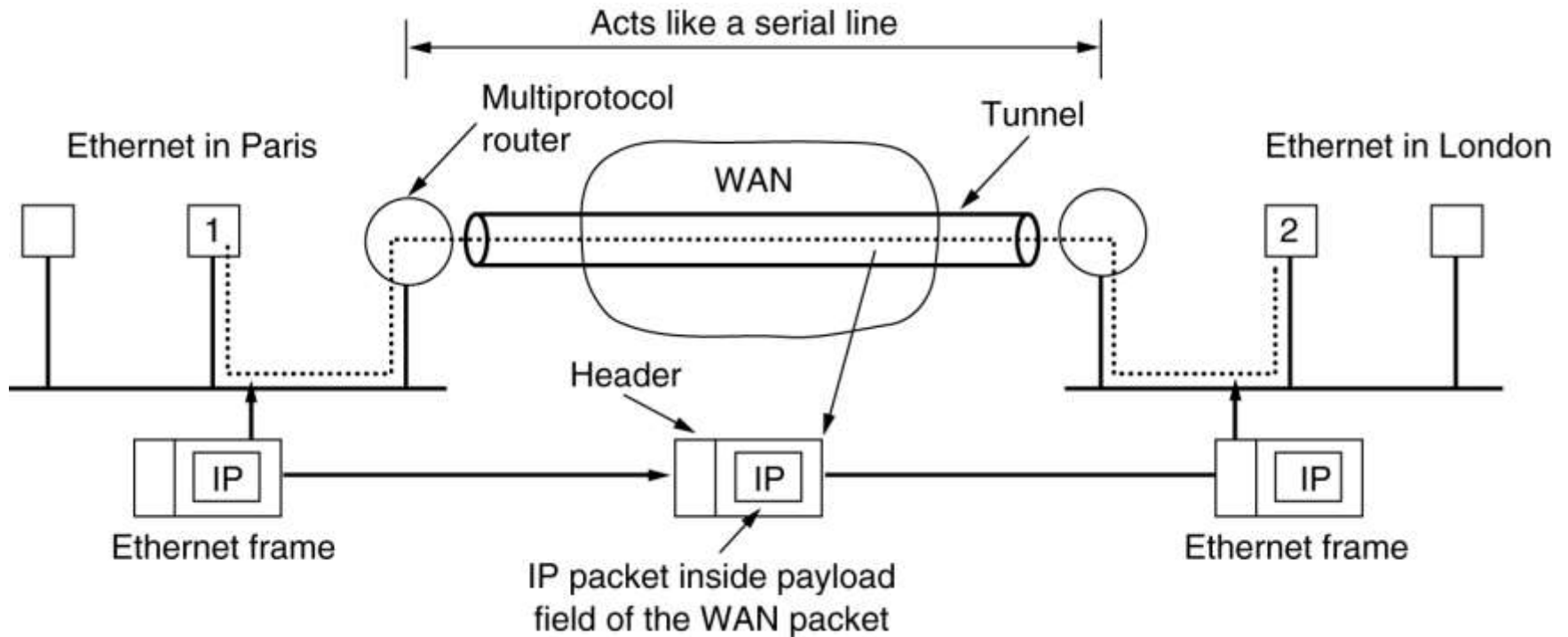
Tunneling is typically encapsulating a network layer packet inside another network layer packet.



Tunneling

- **A typical example :**

Two IP-based Ethernet and a non – IP based WAN between them



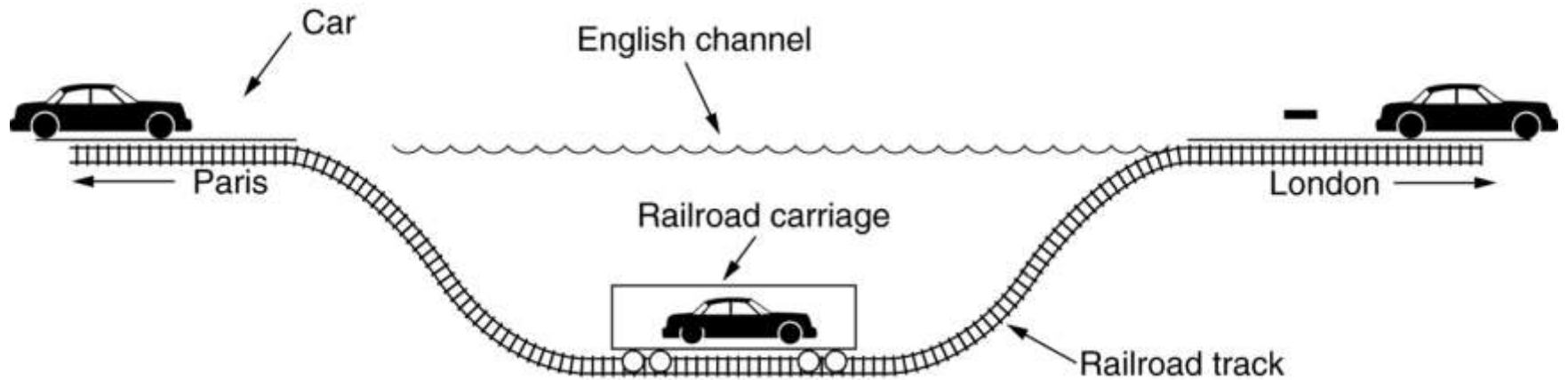
Tunneling

Tunneling Process

1. Host 1 constructs the packet containing the IP address of host 2, inserts it into an Ethernet frame addressed to the Paris multiprotocol router.
2. When the multiprotocol router gets the frame, it removes the IP packet, inserts it in the payload field of the WAN network layer packet.
3. WAN network layer packet is sent to the other router
4. When it gets there, the London router removes the IP packet and sends it to host 2 inside an Ethernet frame

Tunneling

An analogy

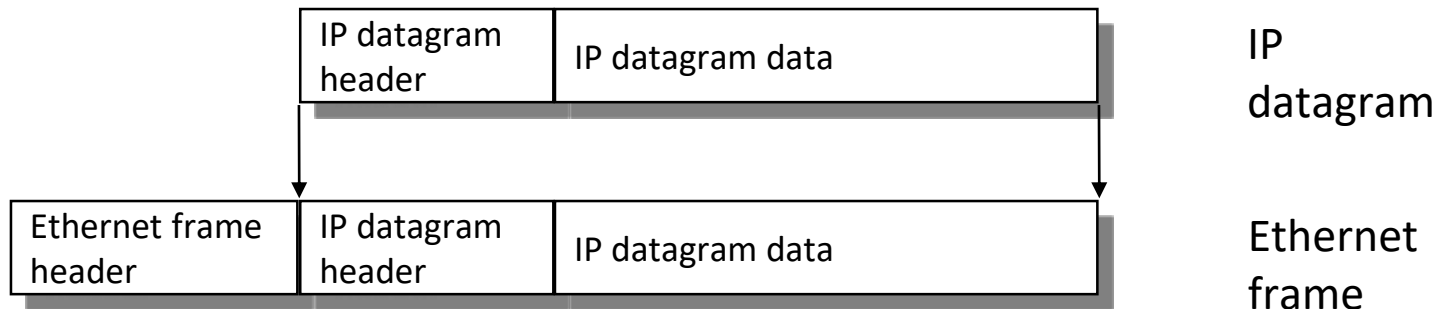


when the car hits the English Channel, it is loaded into a high-speed train and transported to England through the Tunnel

The Internet – Global internet

IP datagram has 2 parts :

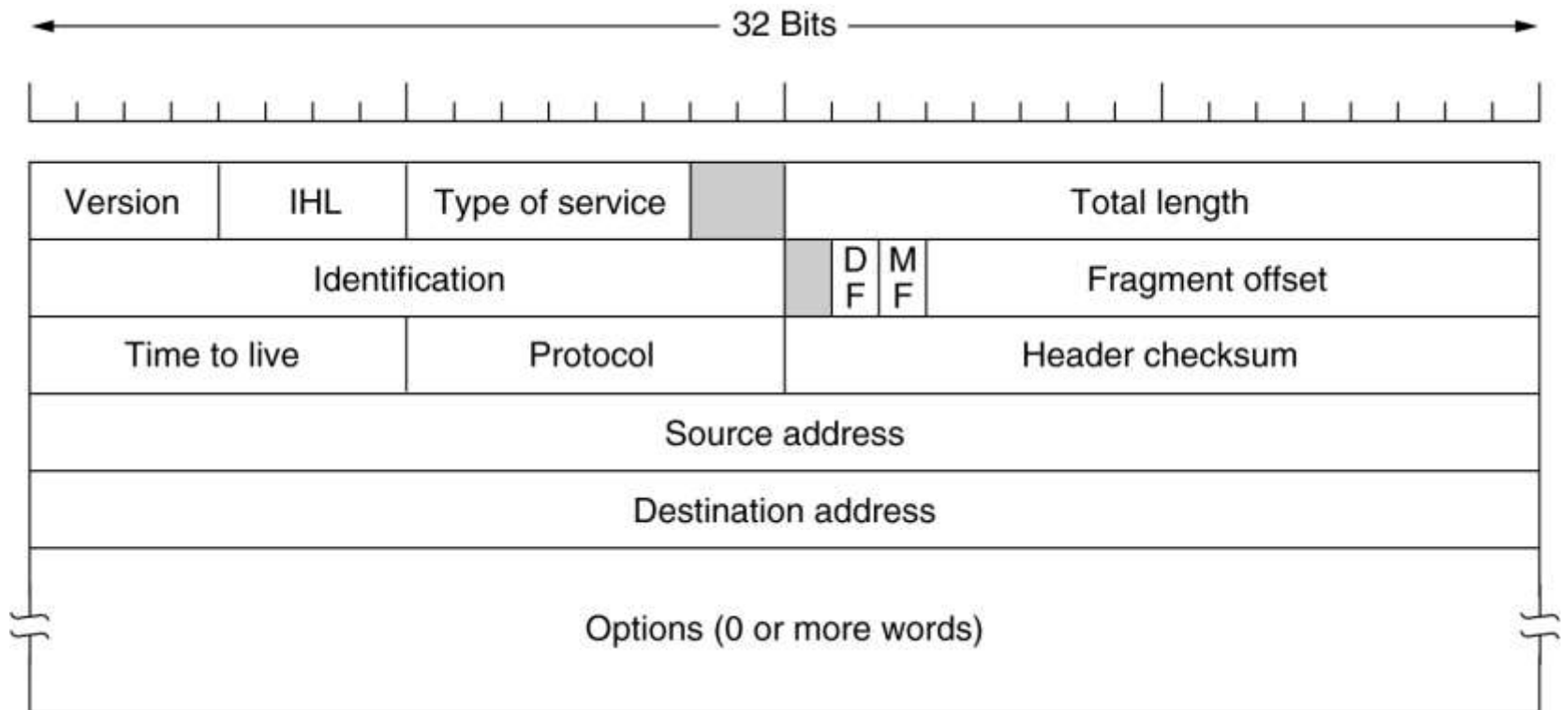
1. IP header
2. IP data



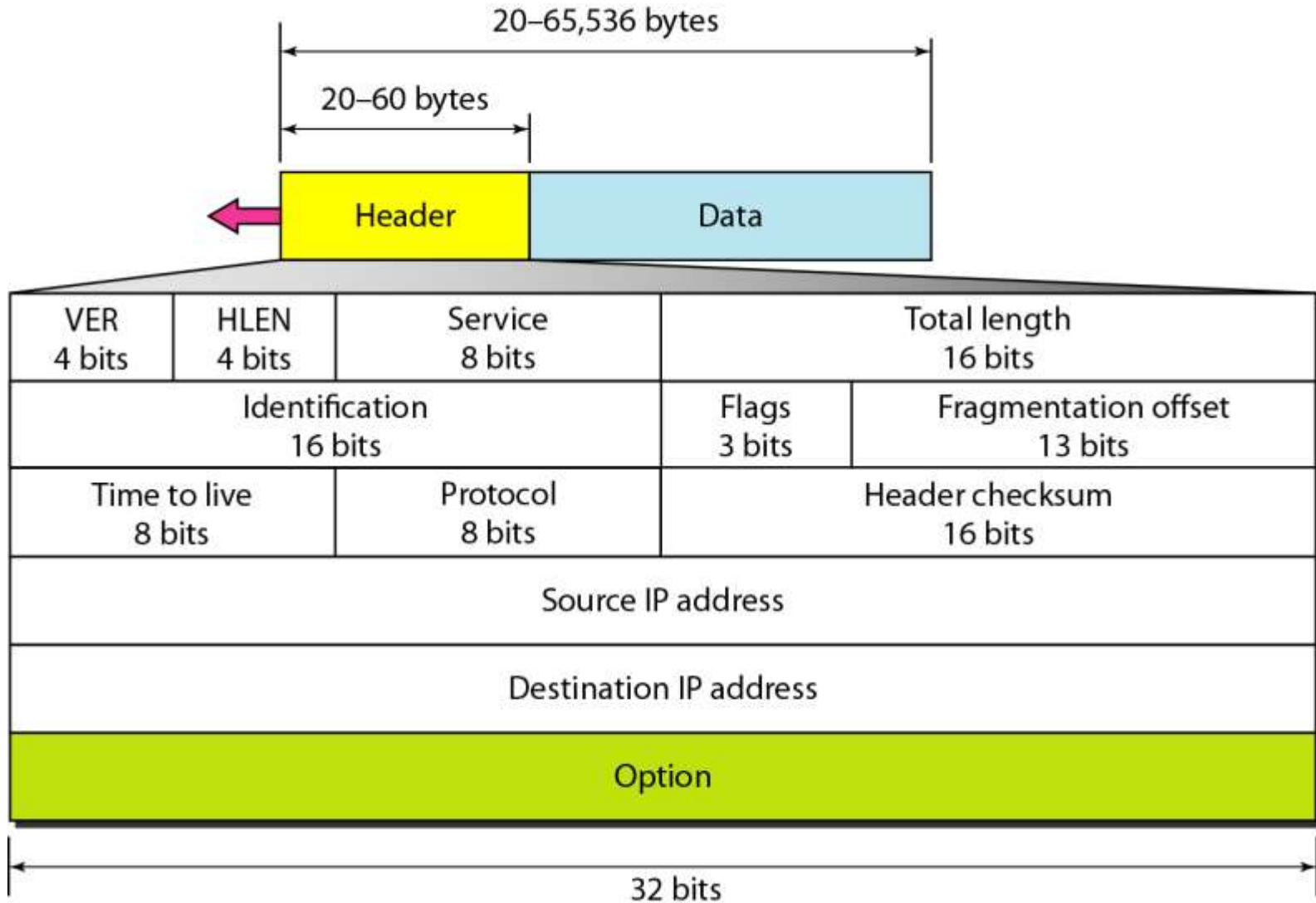
Encapsulating IP datagram in Ethernet

The IP Header

The header has a 20-byte fixed part and a variable length optional part. Transmitted in big-endian order



The IP Header



The IP Header

Version Number (4 bits):

- Specifies the IP protocol version of the datagram
- The router first looks at the version number to determine how to interpret the remainder of the IP datagram
- Different versions of IP use different datagram formats
- The "current" version of IP, is IPv4,

The IP Header

Header Length (IHL) (4 bits):

- Specifies the length of the header in 32-bit words
- When there are no options,, the header is 5 words (20 bytes) long
- The maximum value of this 4-bit field is 15
 $15 \times 4 = 60$ bytes header length
 $60 - 20 = 40$ bytes is the maximum options length

(Most IP datagrams do not contain options)

The IP Header

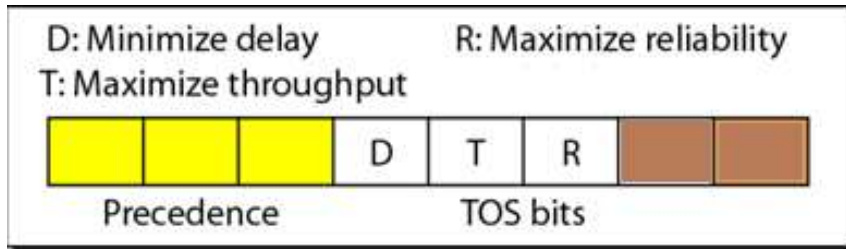
TOS (type of service) : 8 bits

- Basic function is to allow packets to be treated differently based on application needs.
- Originally, the first 6-bit field contained :
 - 3-bit Precedence field (0-7)
 - 3 flags, D, T, and R.
{Delay, Throughput, Reliability}.

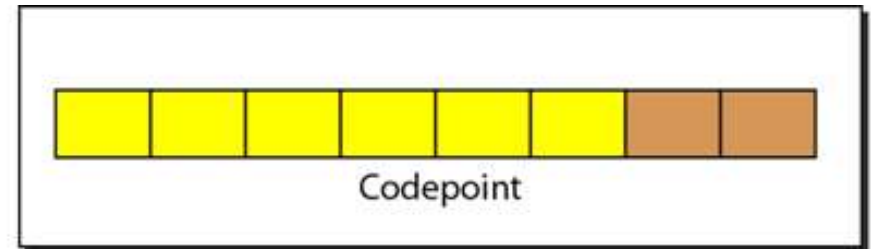
The IP Header

TOS (type of service) : 8 bits

IETF has changed the field slightly to accommodate differentiated services.



Service type



Differentiated services

The IP Header

Total Length : 16 bits

- The total length of the IP datagram (header + data) measured in bytes
- theoretical maximum size of the IP datagram is 65,535 bytes
- In practice datagrams are rarely greater than 1500 bytes

The IP Header

Identifier, Flags, Fragmentation Offset :

- **16 bit identifier** determines which datagram a newly arrived fragment belongs to.
- All the fragments of a datagram contain the same Identification value

The IP Header

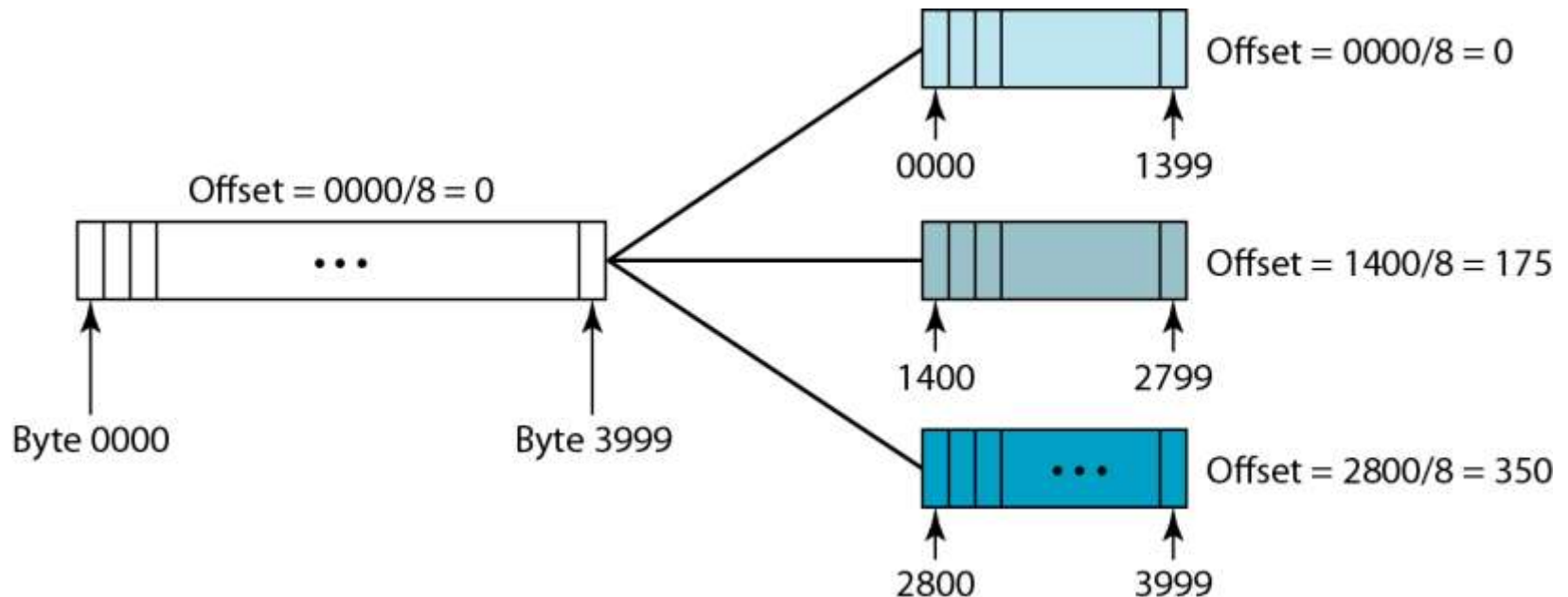
Identifier, Flags, Fragmentation Offset :

- **13 bit The Fragment offset** tells where in the current datagram this fragment belongs.
- Maximum of 8192 fragments per datagram
- All fragments except the last one in a datagram must be a multiple of 8 bytes

The IP Header

Identifier, Flags, Fragmentation Offset :

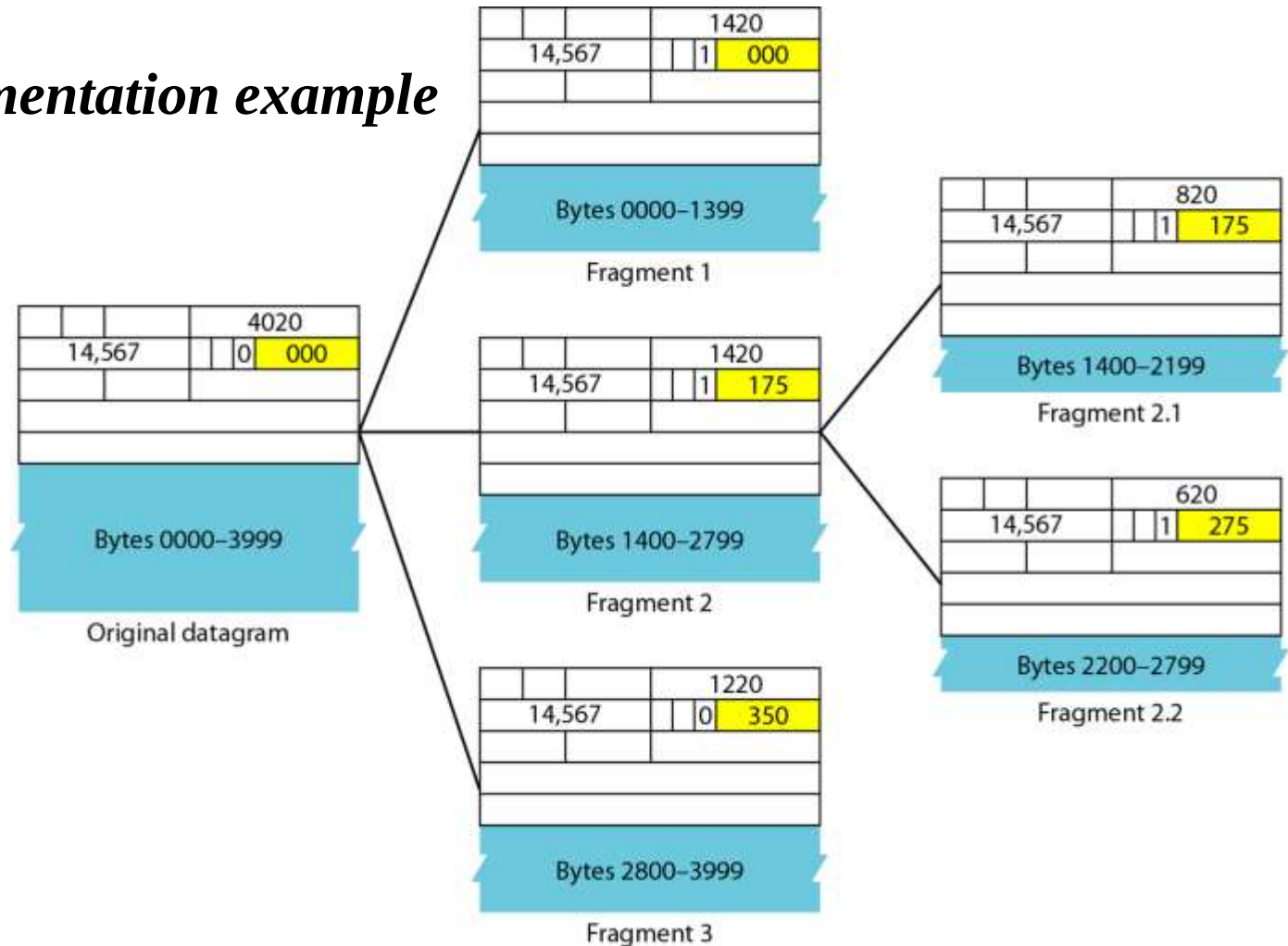
- All fragments except the last one in a datagram must be a multiple of 8 bytes



The IP Header

Identifier, Flags, Fragmentation Offset :

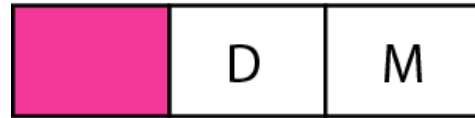
Detailed fragmentation example



The IP Header

Identifier, Flags, Fragmentation Offset :

- **3 bit Flags:**



D: Do not fragment
M: More fragments

- First bit is unused
- 2nd Bit is DF = Don't Fragment
 - Datagram must not be fragmented.
- 3rd bit MF = More Fragments
 - All fragments except the last one have this bit set
 - Allows destination to decide when to do reassembly

The IP Header

Time to live (8 bits)

- Limits packet lifetimes (a maximum lifetime of 255 sec)
- Decrement by one each time the datagram is processed by a router.
- If the TTL field **reaches** 0, the datagram must be dropped

(to ensure that datagrams do not circulate forever in the network)

The IP Header

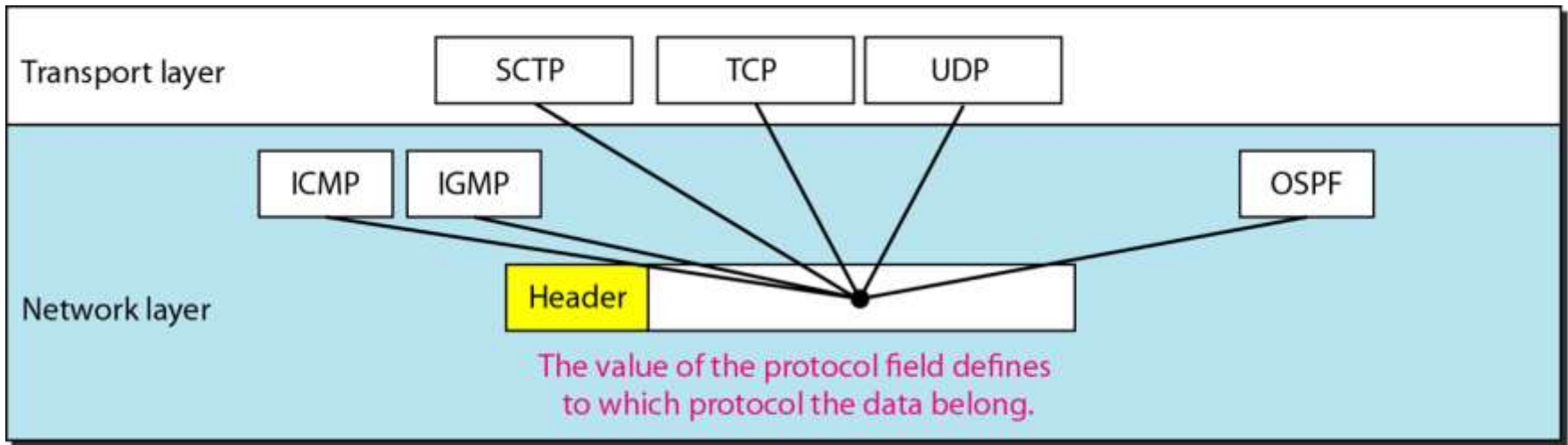
The Protocol field (8 bits)

- Identifies the transport layer protocol to which this IP packet should be passed.
- There are values defined for TCP (6), UDP (17), and many other protocols that may sit above IP

(For a listing of all possible numbers, see [RFC 1700]

The IP Header

The Protocol field (8 bits)



*Protocol
values*

<i>Value</i>	<i>Protocol</i>
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF

The IP Header

Header Checksum (16 bits):

- Helps a router in detecting bit errors in a received IP datagram.
- Computed by treating each 2 bytes in the header as a number and summing these numbers using 1's complement arithmetic.
- then take the 1's complement of this sum (known as the Internet checksum)

The IP Header

Source and Destination IP Address:

- carry the 32 bit IP address of the source and final destination for IP datagram.
- The source address is required to allow recipients to decide if they want to accept the packet and to enable them to reply.

The IP Header

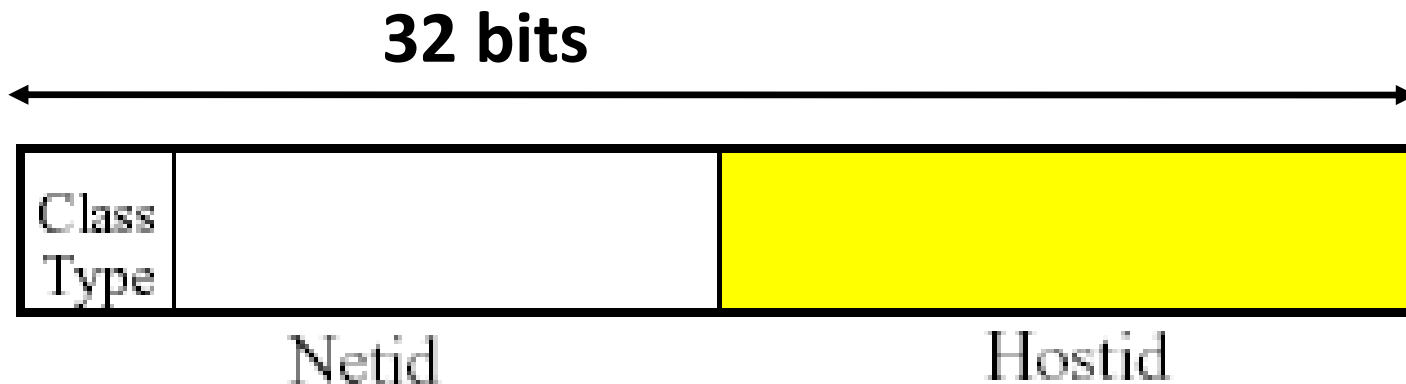
Options (variable length):

- The optional options fields allows an IP header to be extended.
- Originally, five options were defined

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

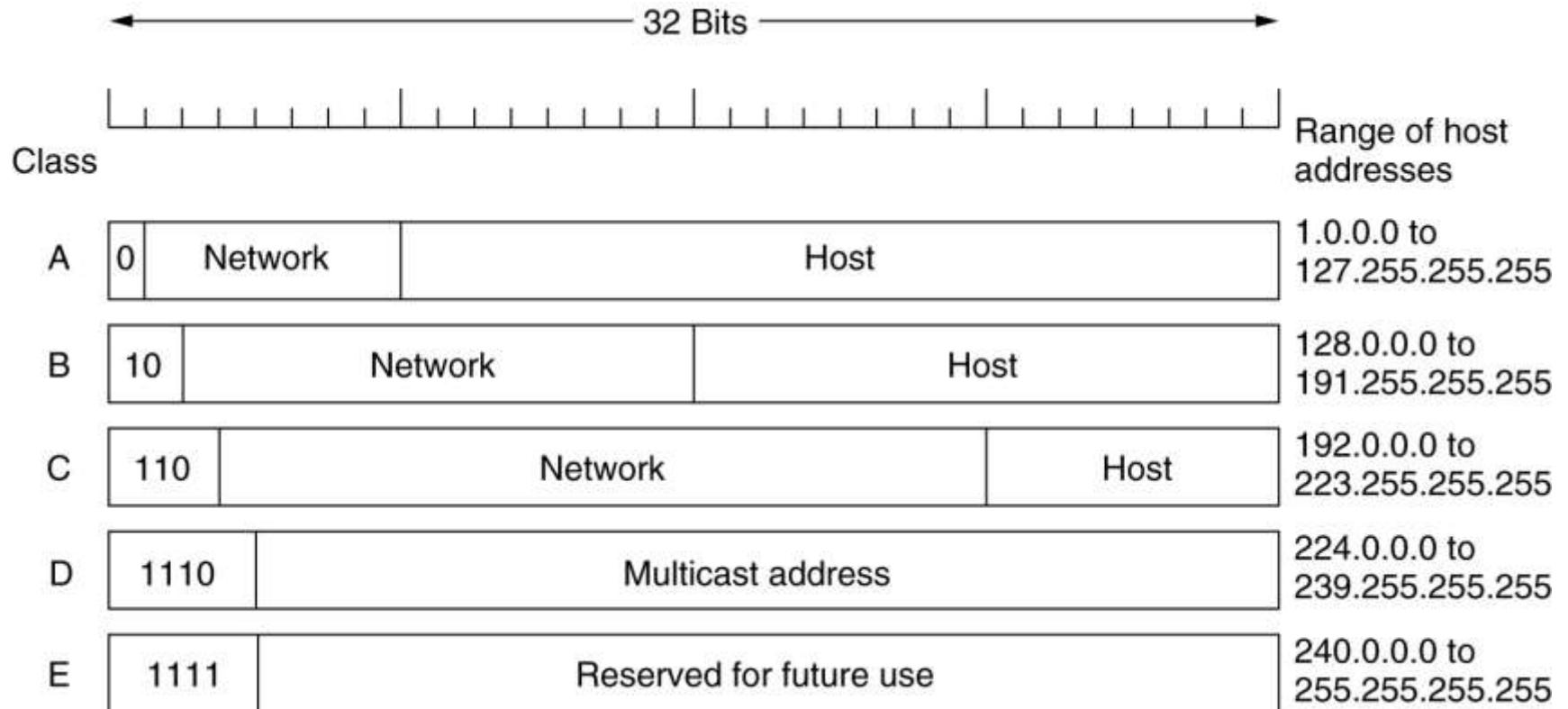
IP Addresses

- IP Address has two parts:
 - Network Part (or Prefix) – high order bits (left)
 - describes which physical network
 - assigned by ICANN or an ISP
 - Host Part – low order bits (right)
 - describes which host on that network
 - Assigned by Network administrators



IP Addresses

- IP addresses are divided into the five categories
 - 32 bit global internet address
 - Network part and host part



This allocation is called **classful addressing**.

IP Addresses

(Network bits)

(Host bits)

Address Class	Bits In Prefix	Maximum Number of Networks	Bits In Suffix	Maximum Number Of Hosts Per Network
A	7	128	24	16777216
B	14	16384	16	65536
C	21	2097152	8	256

- Each network number must be unique
- An organization obtains network numbers from ISPs
- ISPs coordinate with a central organization, the **Internet Corporation for Assigned Names and Numbers (ICANN)**

IP Addresses

➤ **Class A**

➤ 0.0.0.0 to 127.255.255.255

UNICAST

➤ **Class B**

➤ 128.0.0.0 to 191.255.255.255

UNICAST

➤ **Class C**

➤ 192.0.0.0 to 223.255.255.255

UNICAST

➤ **Class D**

➤ 224.0.0.0 to 239.255.255.255

MULTICAST

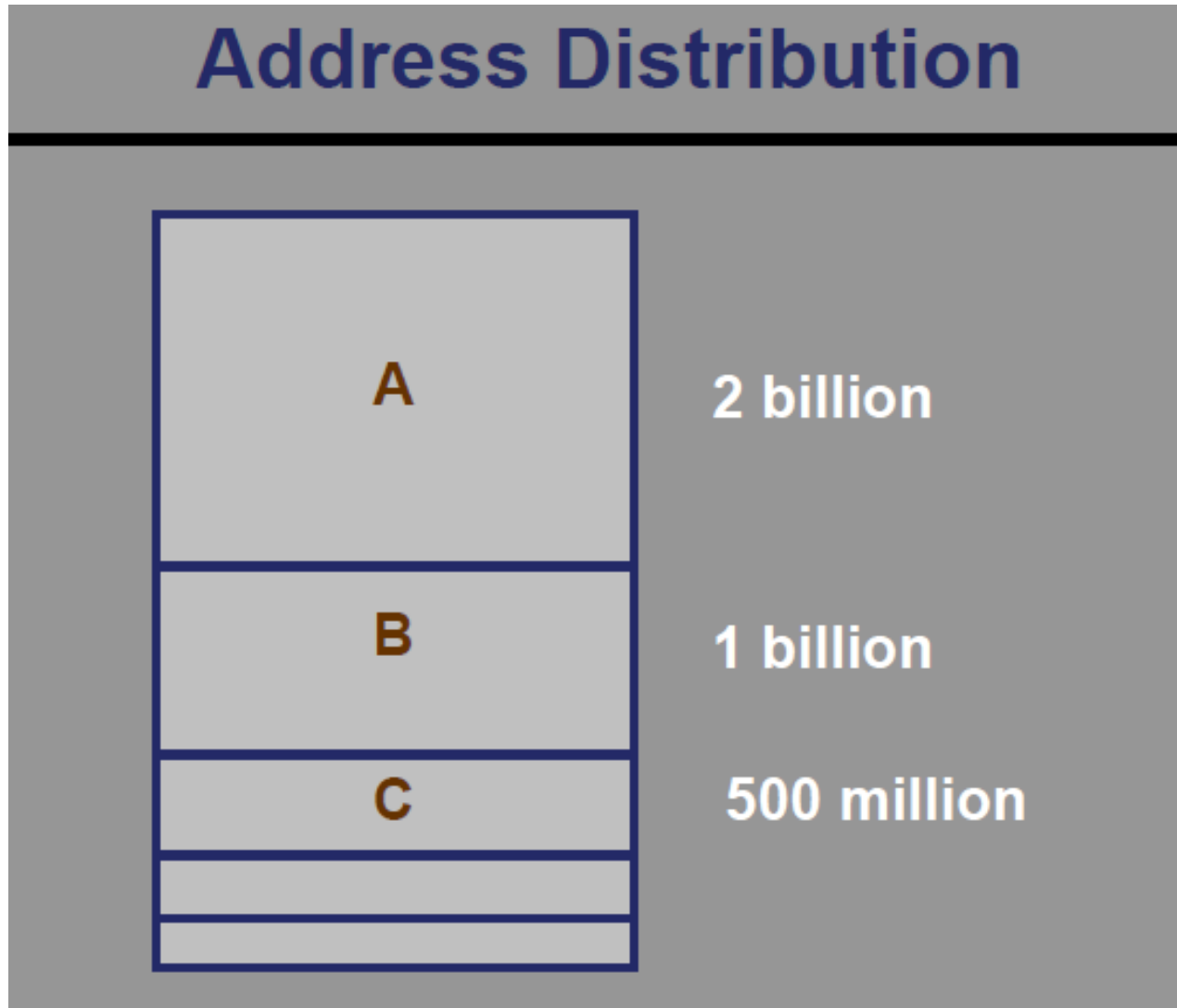
➤ **Class E**

240.0.0.0 to 255.255.255.255

RESERVED

IP address range for each Class

IP Addresses



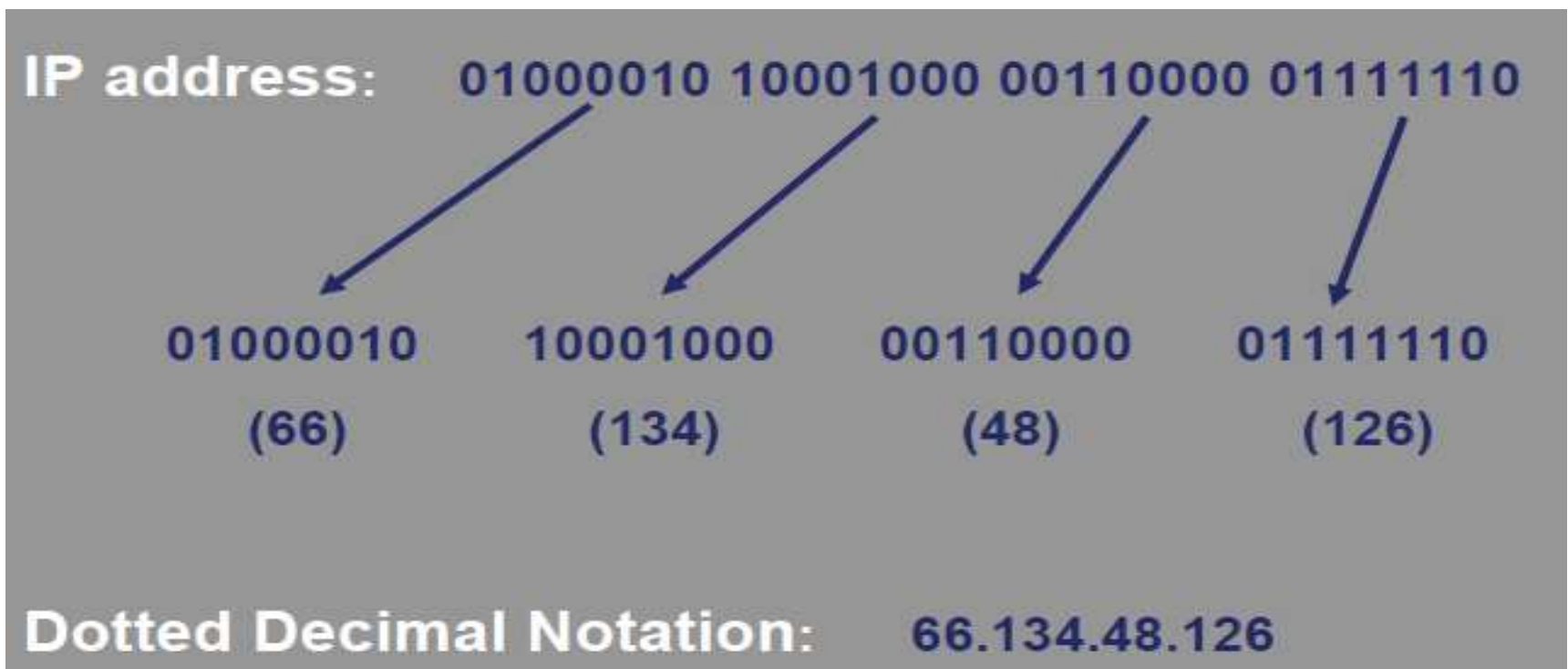
IP Addresses

- IP addresses are self identifying
 - Class of an address can be computed from address itself

First Four Bits Of Address	Table Index (in decimal)	Class of Address
0000	0	A
0001	1	A
0010	2	A
0011	3	A
0100	4	A
0101	5	A
0110	6	A
0111	7	A
1000	8	B
1001	9	B
1010	10	B
1011	11	B
1100	12	C
1101	13	C
1110	14	D
1111	15	E

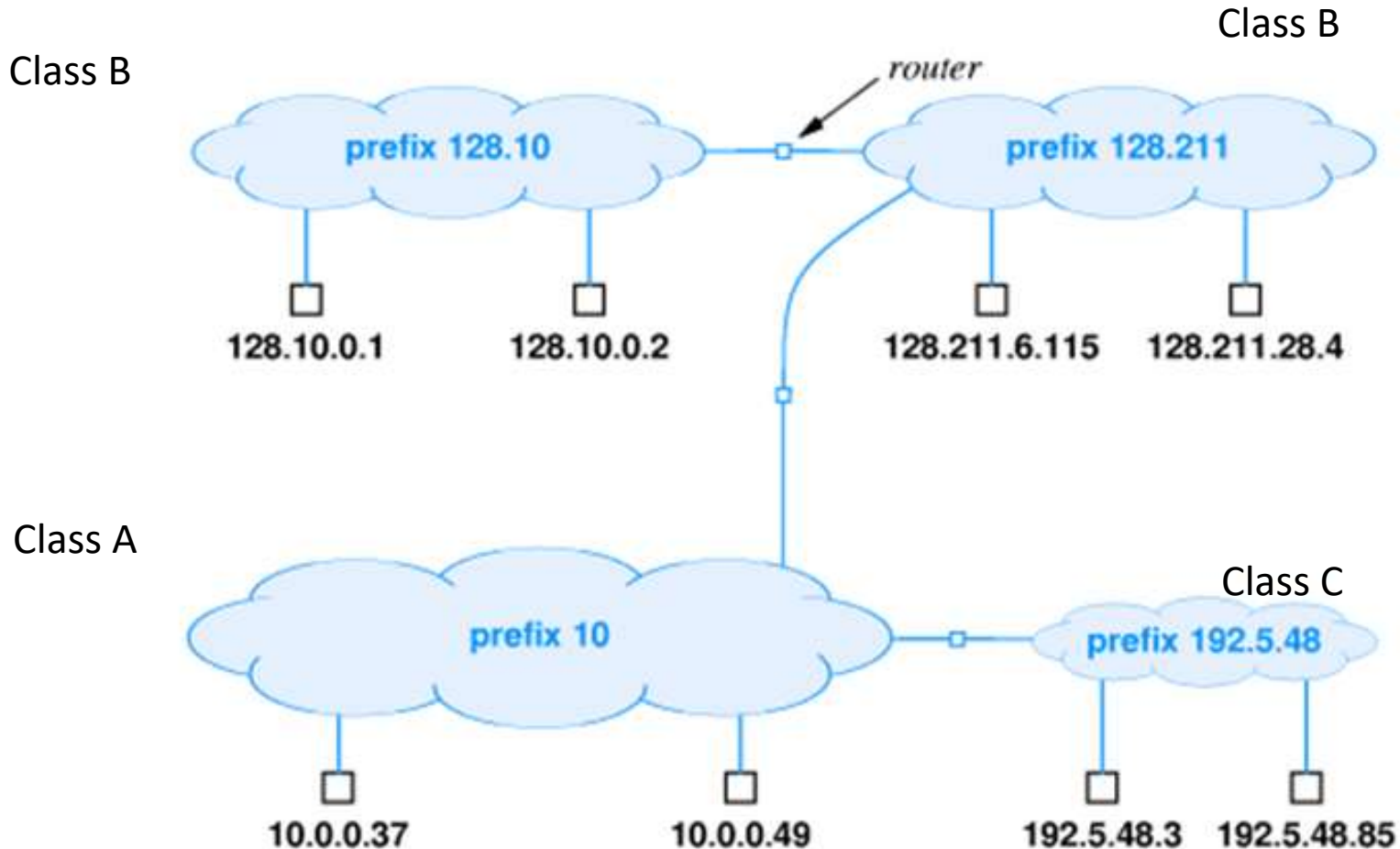
IP Addresses

- Expressed as a dotted-decimal notation W.X.Y.Z, where dots are used to separate each of the four octets of the address.



IP Addresses

- An Addressing Example



Special IP Addresses

- IP defines a set of special address forms that are reserved

0 0																														This host
0 0 . . . 0 0										Host																				A host on this network
1 1																														Broadcast on the local network
Network										1 1 1 1 . . . 1 1 1 1										Broadcast on a distant network										
127					(Anything)																									Loopback

Special IP addresses.

Special IP Addresses

Special IP addresses.

Prefix	Suffix	Type of Address	Purpose
All 0s	All 0s	This host /network	Used during bootstrap
All 0s	host	This Network	A host on this network
Network	All 1s	Direct broadcast	Broadcast on a specified network
All 1s	All 1s	Limited broadcast	Broadcast on a local network
127	any	loop back	testing

IP Addresses - Class A & B

Class A

- Start with binary 0
- 7-bit network - 24-bit host
- 01111111 (127) reserved for loopback
 - Generally 127.0.0.1 is used
- Range 1.x.x.x to 126.x.x.x
 - 10.x.x.x is for private networks
- Few networks - many hosts
- All allocated

Class B

- Starts with binary 10
- 14-bit network, 16-bit host number
 - $2^{14} = 16,384$ class B addresses
 - $2^{16} = 65536$ hosts per network
- Range 128.x.x.x to 191.x.x.x
 - Second octet is also part of the network id.
- All allocated

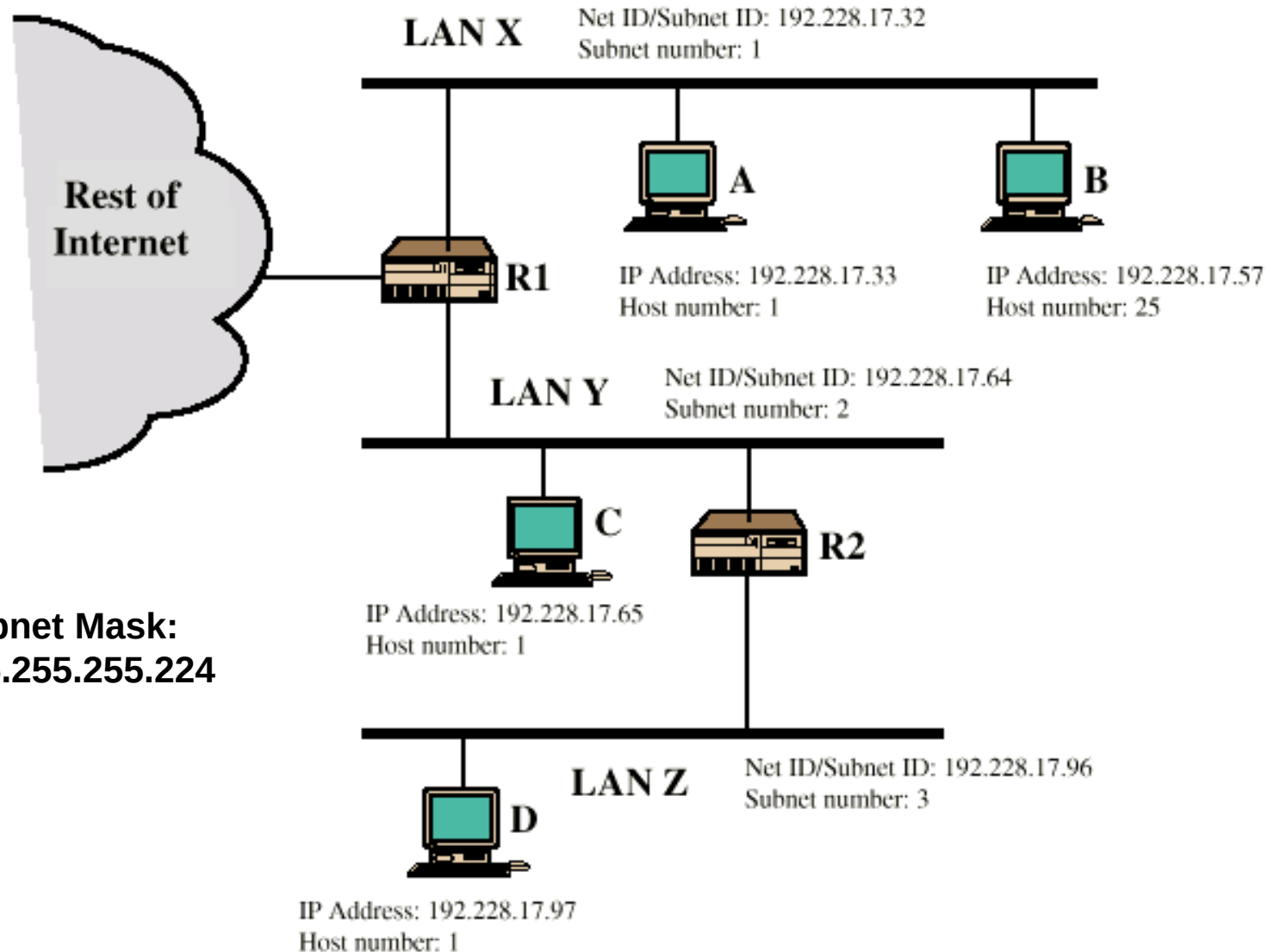
IP Addresses - Class C

- Start binary 110
- Range 192.x.x.x to 223.x.x.x
- Second and third octet also part of network address
- $2^{21} = 2,097,152$ addresses (networks)
- 256 hosts per network
- Nearly all allocated
 - See IPv6

Subnets and Subnet Masks

- Allow arbitrary complexity of internetworked LANs within organization
 - By not having one network class for each LAN within the organization
- Site looks to rest of internet like single network
- Each LAN is assigned a subnet number
- Host portion of address partitioned into subnet number and host number
- Local routers route within subnetted network
- Subnet mask indicates which bits are subnet number and which are host number

Routing Using Subnets



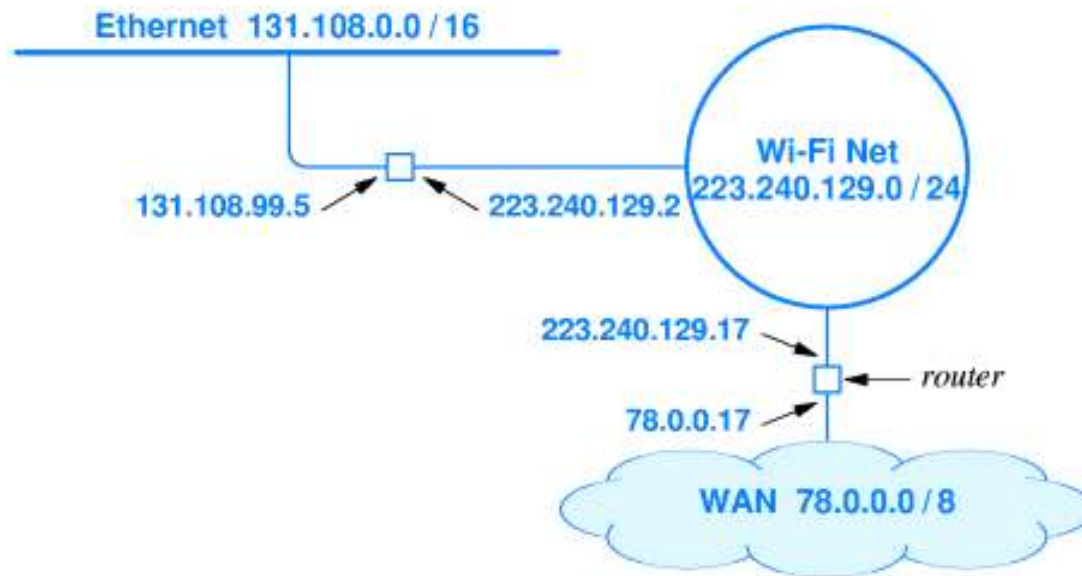
- Subnet Mask:
255.255.255.224

Classless Addresses

- Extension of subnet idea to the whole Internet
- Assigning IP numbers at any size together with a subnet number
- A precaution against exhaustion of IP addresses
- Special notation (CIDR notation)
 - network address/number of 1-bits in the mask
 - e.g. 128.140.168.0/21
 - subnet mask is 255.255.248.0
 - Lowest host address?
 - Highest host address?

Example Network Configuration

- IP address is the address of a connection (not of a computer or router)



IP v6 - Version Number

- IP v 1-3 defined and replaced
- IP v4 - current version
- IP v5 - stream protocol
 - Connection oriented internet layer protocol
- IP v6 - replacement for IP v4
 - Not compatible with IP v4
 - During the initial development it was called IPng (Next Generation)

Driving Motivation to change IP

- Address space exhaustion
 - Two level addressing (network and host) wastes space
 - Growth of networks and the Internet
 - Extended use of TCP/IP
 - e.g. for POS terminals
 - wireless nodes
 - vehicles

IETF started work on new version with following goals:

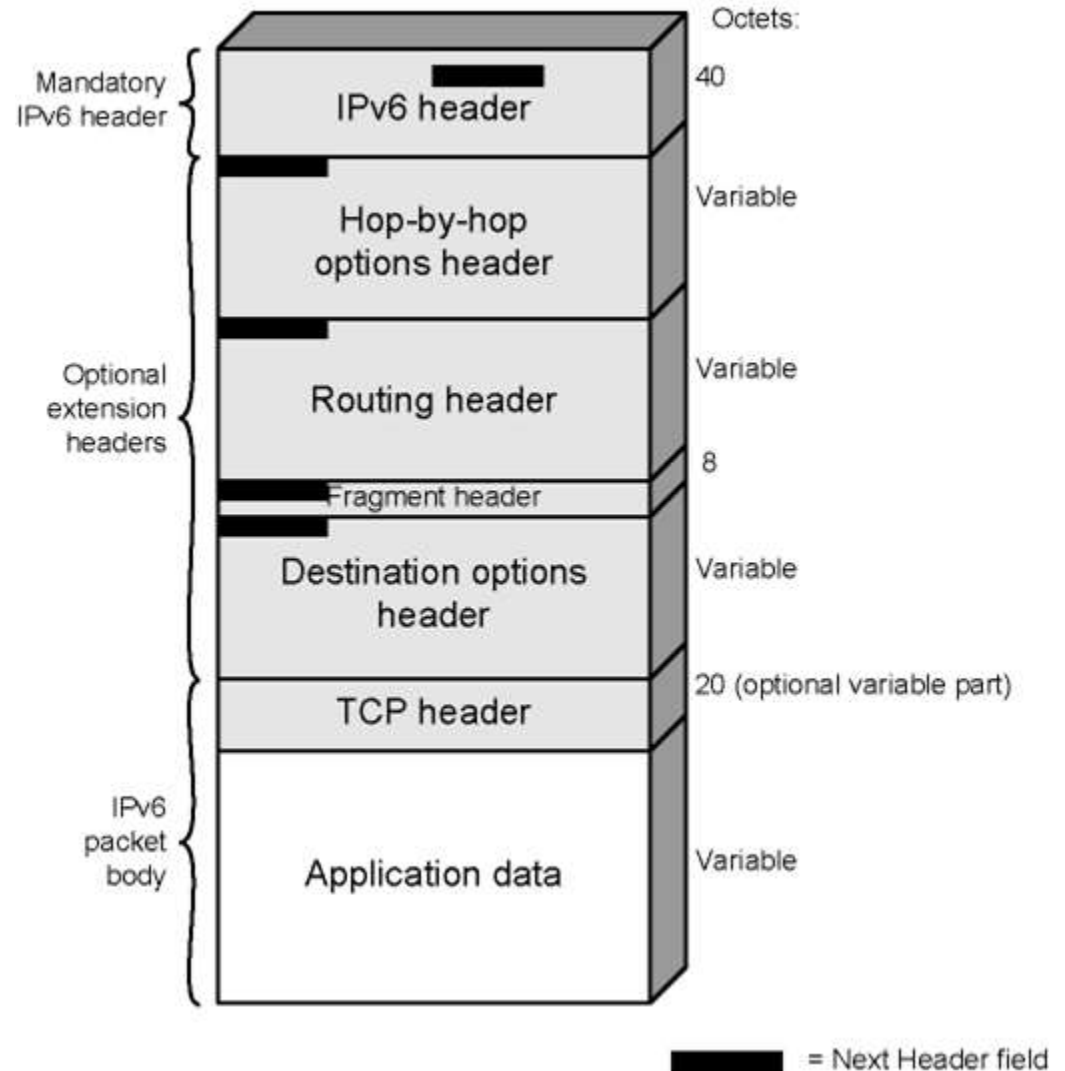
- To support billions of host
- Size of routing table must be reduced
- To allow router to process packets further by simplifying the protocol
- Provide better security mechanism
- More importance must be given to type of service
- Supports multicasting by allowing scopes to be specified
- Mobility for a host to move without changing its address
- Allow the protocol to evolve in future
- Permit the old and new protocols to co-exist for years

IPv6 Enhancements

- Expanded address space
 - 128 bit
 - 6×10^{23} addresses per square meter on earth
- Improved option mechanism
 - Separate optional headers between IPv6 header and transport layer PDU
 - Most are not examined by intermediate routers
 - Improved speed and simplified router processing
 - Easier to extend options
 - Flexible protocol
- Support for resource allocation
 - Labeling of packets for particular traffic flow
 - Allows special handling
 - e.g. real time video

IPv6 Packet with Extension Headers

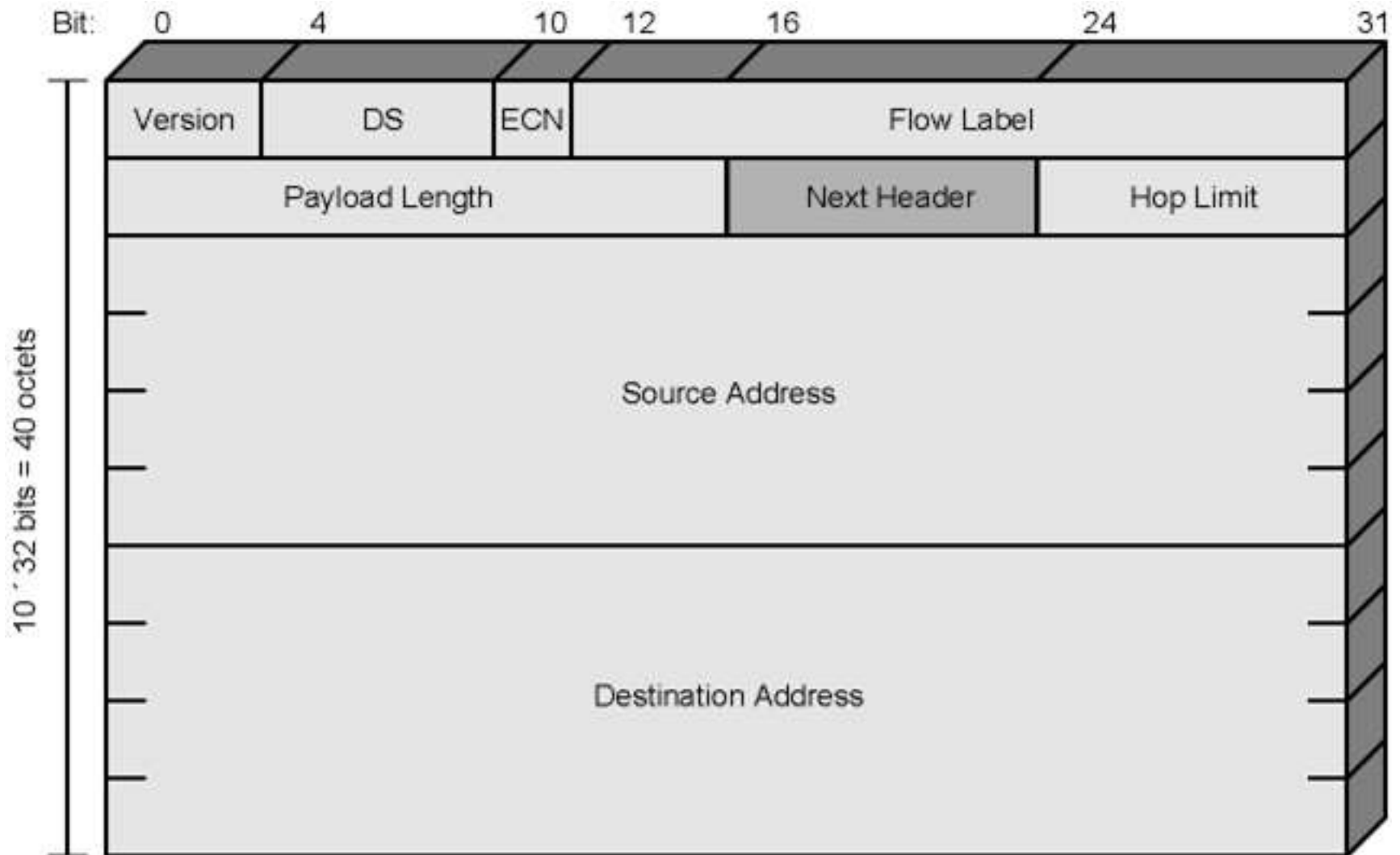
- IPv6 headers + optional extension headers



Extension Headers

- Hop-by-Hop Options
 - special options that require hop-by-hop processing
- Routing
 - Similar to source routing(List of routers to visit)
- Fragment
 - fragmentation and reassembly information
- Authentication
 - Integrity and Authentication
- Encapsulating security payload
 - Privacy and Confidentiality
- Destination options
 - Optional info to be processed at destination node

IPv6 Header



IP v6 Header Fields (1)

- Version
 - 6
- DS/ECN
 - Previously, Traffic Class (Types of Service)
 - Classes or priorities of packet
 - Now interpretation is different as discussed in v4
- Flow Label
 - Identifies a sequence of packets (a flow) that has special handling requirements
- Payload length
 - Includes all extension headers plus user data

IP v6 Header Fields (2)

- Next Header
 - Identifies type of header
 - Extension or next layer up
- Source Address
- Destination address
- Longer header but less number of fields
 - simplifies processing

Flow Label

- Flow
 - Sequence of packets from particular source to particular destination
 - Source desires special handling by routers
 - Uniquely identified by source address, destination address, and 20-bit flow label
- Router's view
 - Sequence of packets that share some attributes affecting how packets handled
 - Path, resource allocation, discard needs, accounting, security
 - Handling must somehow be arranged
 - Negotiate handling ahead of time using a control protocol

Differences Between v4 and v6 Headers

- No header length (IHL) in v6
 - header is of fixed length in v6
- No Protocol info in v6
 - next header field will eventually point to the transport layer PDU
- No fragmentation related fields in v6 base header
 - fragmentation is an extension header
- No checksum in v6
 - rely on reliable transmission medium and checksums of upper and lower layers

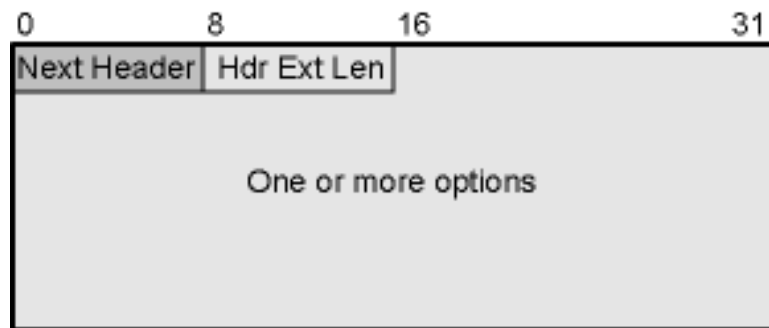
IPv6 Addresses

- 128 bits long
- Assigned to interface
 - An interface may have multiple addresses
- network/host id parts
 - arbitrary boundary
 - like CIDR addresses in v4
- Multilevel hierarchy
 - ISP - Organization - Site - ...
 - Helps faster routing due to aggregation of IP addresses
 - Smaller routing tables and faster lookup
- IPv4 addresses are mapped into v6 addresses
- Three types of address

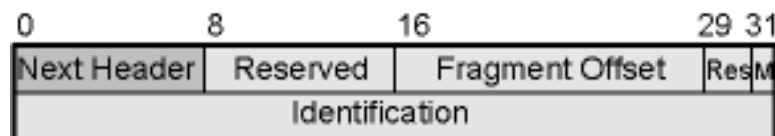
Types of address

- Unicast
 - One address is assigned to a single interface
- Anycast
 - Set of computers (interfaces) that share a single address
 - Delivered to any one interface
 - the “nearest”
- Multicast
 - Set of interfaces
 - Delivered to all interfaces identified

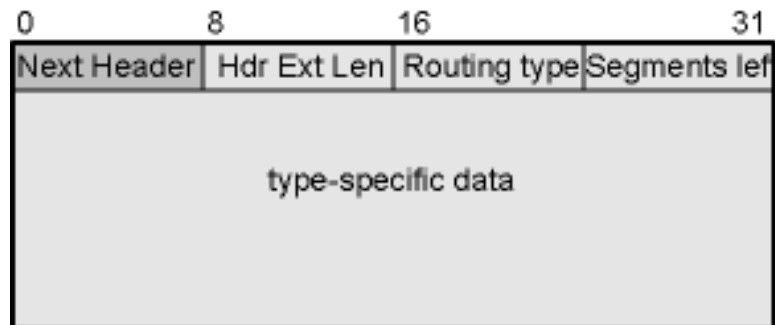
IPv6 Extension Headers



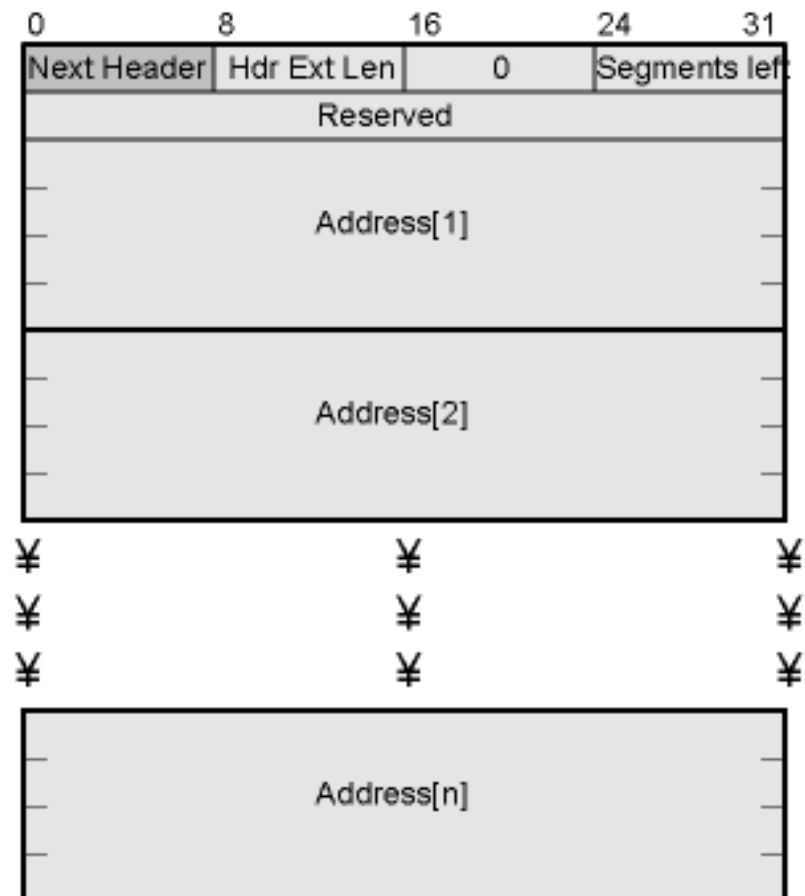
(a) Hop-by-hop options header,
destination options header



(b) Fragment header



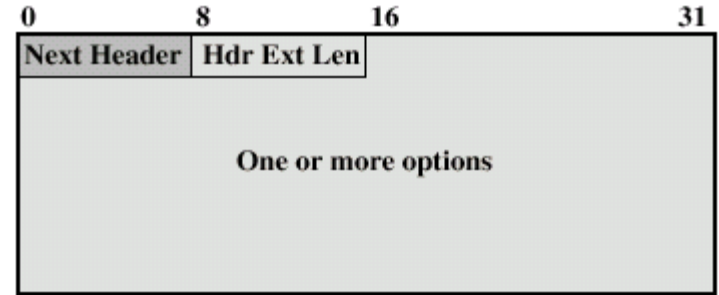
(c) Generic routing header



(d) Type 0 routing header

Hop-by-hop Options

- Next header
- Header extension length
- Options
 - type, length, option data
 - type also says what should router do if it does not recognize the option
 - Pad1 / Pad N
 - Insert one/N byte(s) of padding into Options area of header
 - Ensure header is multiple of 8 bytes
 - Jumbo payload (Jumbogram)
 - for over $2^{16} = 65,535$ octets - up to 2^{32} octets
 - for large video packets
 - Router alert
 - Tells the router that the content of packet is of interest to the router
 - Provides support for Resource Reservation Protocol (RSVP)
 - Traffic control purposes



(a) Hop-by-hop options header;
destination options header

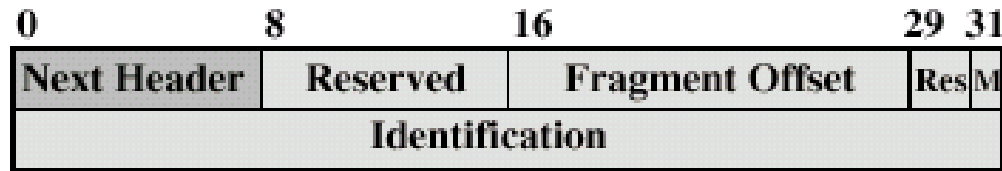
Destination Options

- Same format as Hop-by-Hop options header

Fragment Header

- Fragmentation only allowed at source
- No fragmentation at intermediate routers
- Node must perform path discovery to find smallest MTU (max. transmission unit) of intermediate networks
 - iterative process
- Source fragments to match MTU
- Otherwise limit to 1280 octets
 - 1280 is the minimum supported by each network

Fragment Header Fields

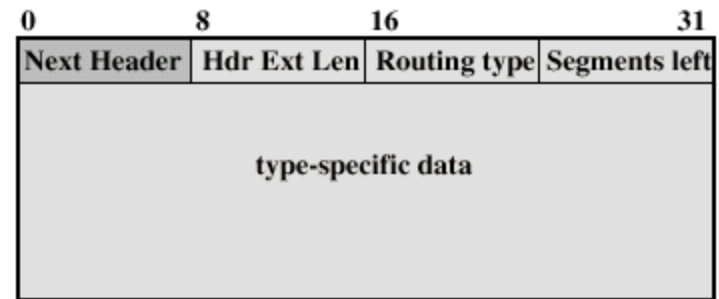


(b) Fragment header

- Next Header
- Fragmentation offset
 - as in v4
- More flag
 - as in v4
- Identification
 - as in v4

Routing Header

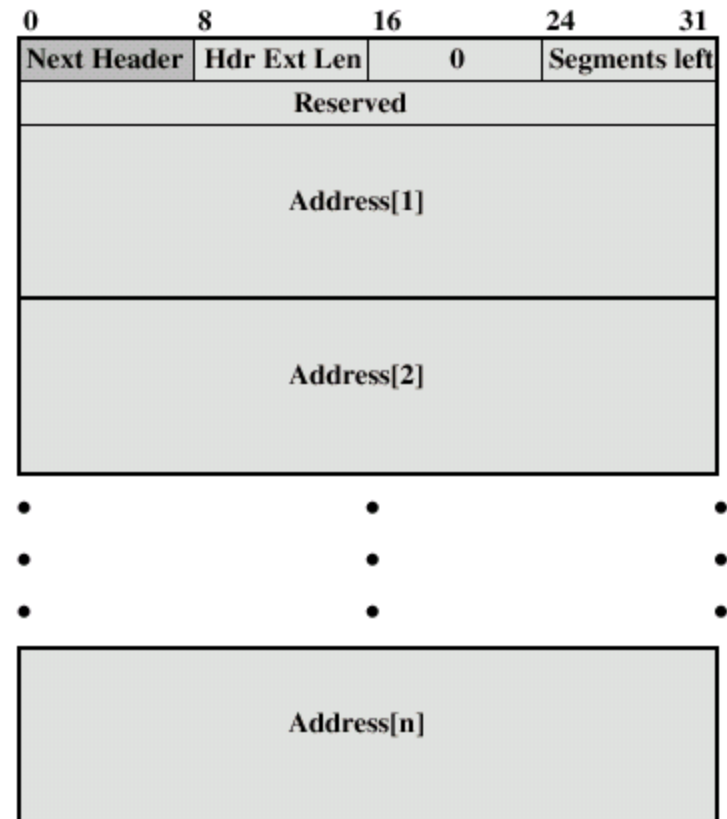
- Source routing method of IPv6
- List of one or more intermediate nodes to be visited(ip addresses of intermediate nodes in sequence)
- Next Header
- Header extension length
- Routing type -0 ,..
- Segments left
 - i.e. number of nodes still to be
 - to be visited when it hits 0 , packet might be near to destination hence would take best route thereon.



(c) Generic routing header

Routing Header

- Type 0 routing
 - The only one defined in RFC 2460
- Base header contains the address of next router
- Router examines the routing header and replaces the address in the base header



(d) Type 0 routing header

Migration to IPv6

- Not an overnight operation
 - lots of investments in v4 networking equipment
 - may take 10s of years
- isolated v6 islands
 - communicating via tunnels
- eventually those islands will get larger and merge

IPv4 and IPv6 Security

- Section 16.6
- IPSec
- Security within the IP level
 - so that all upper level applications will be secured
 - authentication and encryption

The authentication header provides a mechanism by which the receiver of a packet can be sure of who sent it. I.e it is coming from a legitimate person(not an imposter). The Encrypted security payload makes it possible to encrypt the contents of a packet so that only the intended recipient can read it. These headers use cryptographic techniques to accomplish their mission.

IPSec Scope

- Authentication header (AH)
 - authentication only
- Encapsulated Security Payload (ESP)
 - encryption + optional authentication
- Key exchange
 - Oakley, IKE, ISAKMP
- RFC 2401,2402,2406,2408,2409

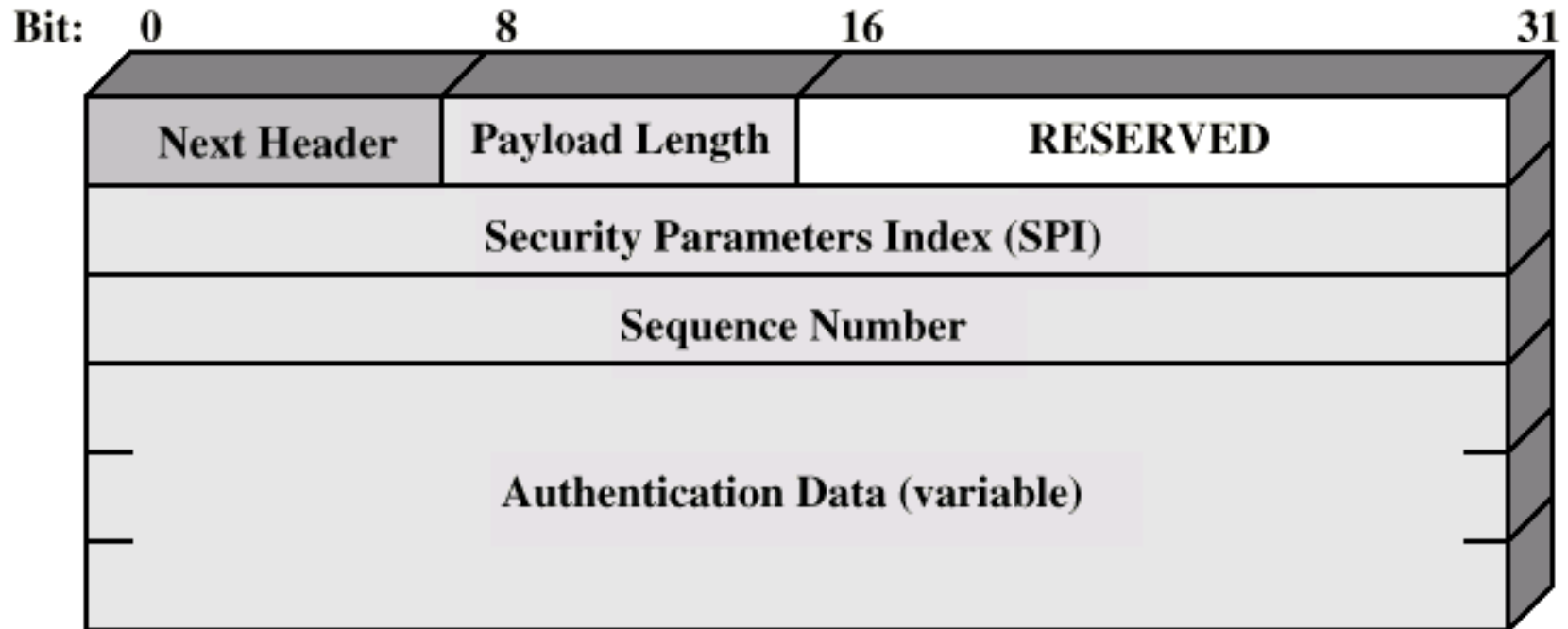
Security Association

- Identifies security relationship between sender and receiver
- Details are at local databases

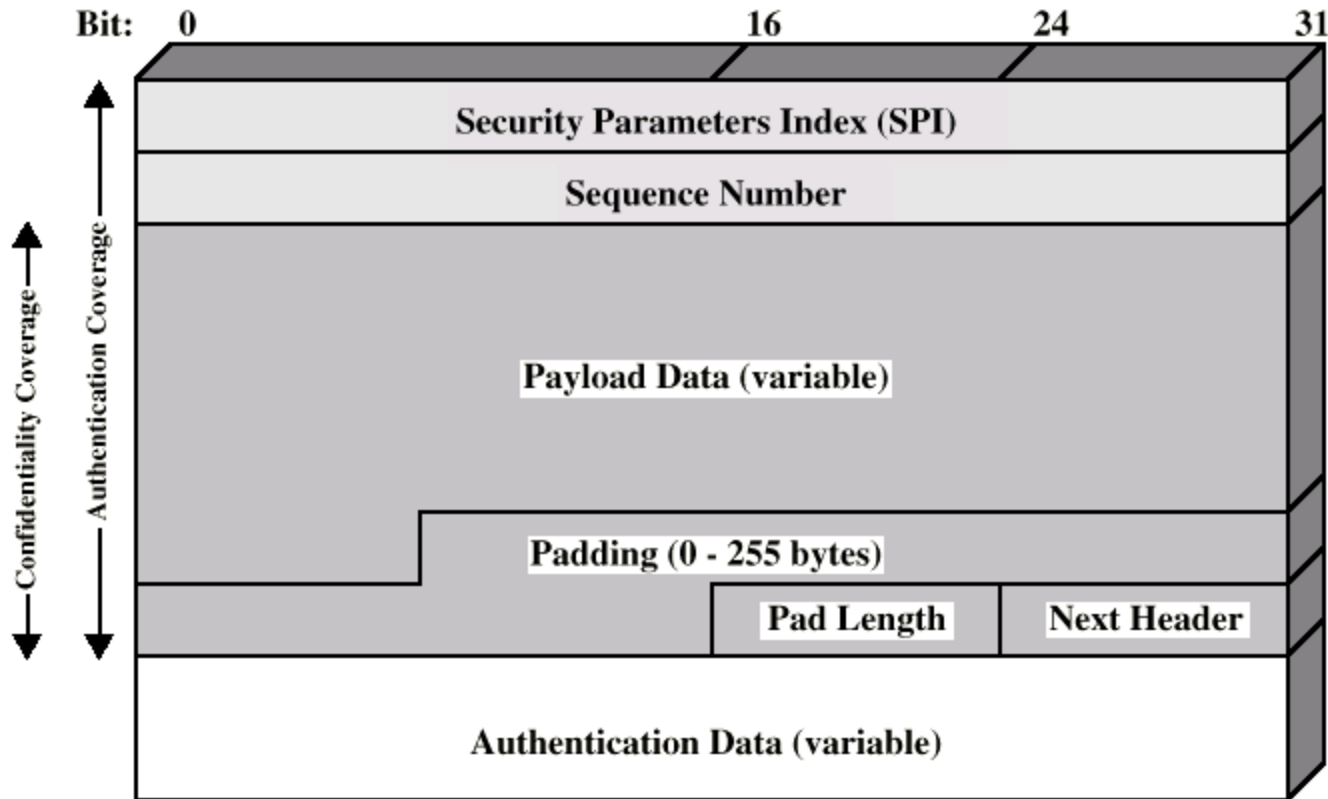
Transport and Tunnel Modes

- Transport mode
 - Protection for upper layer protocol
 - Protection extends to the payload of IP packet
 - generally headers are not included
 - End to end between hosts
- Tunnel mode
 - Protection for the entire IP packet
 - Entire packet treated as payload for outer IP “packet”
 - No routers examine inner packet
 - mostly for router to router connection
 - VPNs (Virtual Private Networks) are constructed in this way

Authentication Header



ESP Packet



- Next Header identifies the first header in the payload